Physical Chemistry of Materials

Lecture 03

Phase Transitions

Last Class

Equilibrium

- An isolated system is in equilibrium when its properties remain constant with time

- Mechanical equilibrium: constant p and V
- Diffusive equilibrium: constant n
- Thermal equilibrium: constant T

Last Class

First Law of Thermodynamics

"The internal energy of an isolated system is constant."

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$\mathrm{d}U = \delta q + \delta w$$

small change of U | small q | small w

Entropy

$$\mathrm{d}S = \frac{\delta q_{\mathrm{rev}}}{T}$$

- Index of exhaustion
- Second law of thermodynamics

$$\mathrm{d}S \geq 0 \text{ (isolated system)}$$

  - $\mathrm{d}S = 0$: reversible change
  - $\mathrm{d}S > 0$: irreversible change

- Third law of thermodynamics: $\lim_{T \to 0} S = 0$

Last Class

Fundamental Equation

From the first law,

$$\mathrm{d}U = \delta q + \delta w$$

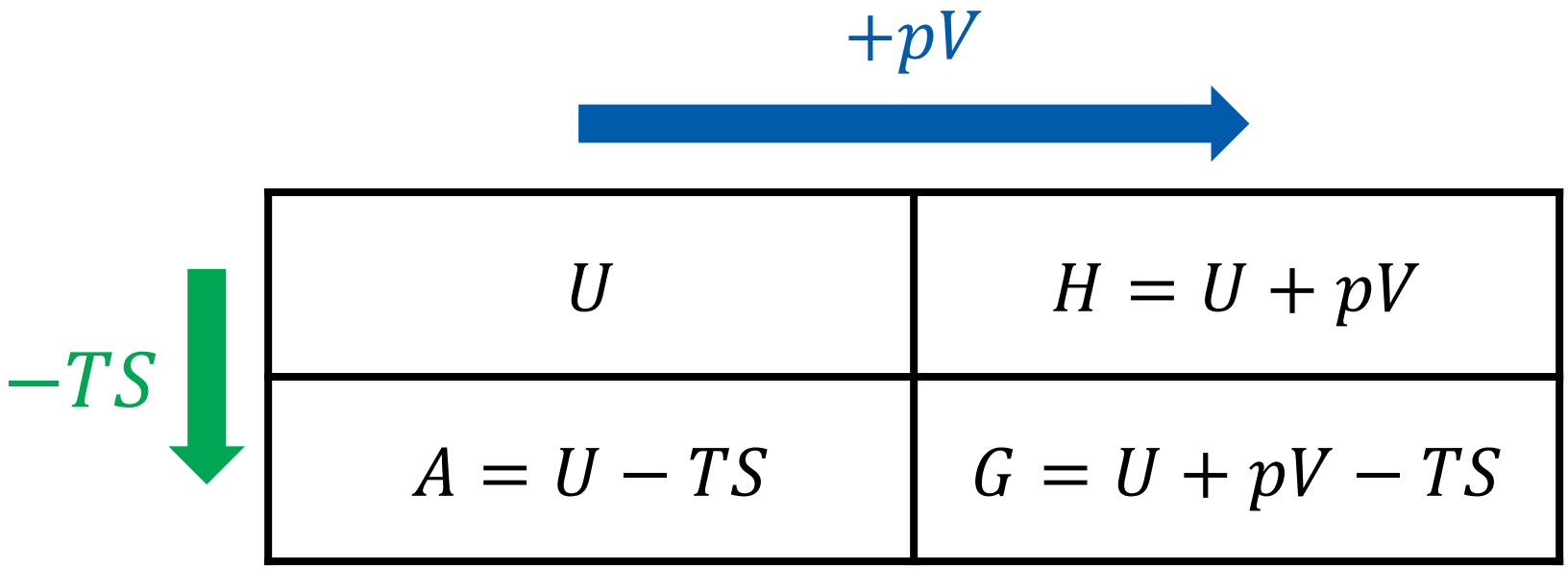
For a reversible change,

$$\mathrm{d}U = \delta q_{\mathrm{rev}} + \delta w_{\mathrm{rev}} = T\mathrm{d}S - p\mathrm{d}V$$

Since all variables are state functions,

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V \text{ (in general)}$$

Thermodynamic Potentials



Natural Independent Variables

- Internal energy $U = U(S, V)$

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Enthalpy $H = U + pV = H(S, p)$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

Two Systems in Equilibrium

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: $\mathrm{d}V$
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $\mathrm{d}S = \delta q/T$
- Diffusive equilibrium: ?
  - Conjugate extensive variable: $\mathrm{d}n$

Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: $\mathrm{d}n$

- (Modified) differentials of thermodynamic potentials

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + \mu\mathrm{d}n$$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}n$$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

Three Independent Variables

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

This implies that $U = U(S, V, n)$, which leads to

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_{V,n} \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_{S,n} \mathrm{d}V + \left(\frac{\partial U}{\partial n}\right)_{S,V} \mathrm{d}n$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \qquad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \qquad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

Meaning of μ

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

Meaning of μ

$$g(T,p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n}\right)_{T,p}$$

However,

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

gives

$$\mu = \left(\frac{\partial G}{\partial n}\right)_{T,p} = g(T,p) = \frac{G}{n}$$

(molar Gibbs energy)

Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$

Two Systems in Equilibrium

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: $\mathrm{d}V$
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $\mathrm{d}S = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: $\mathrm{d}n$

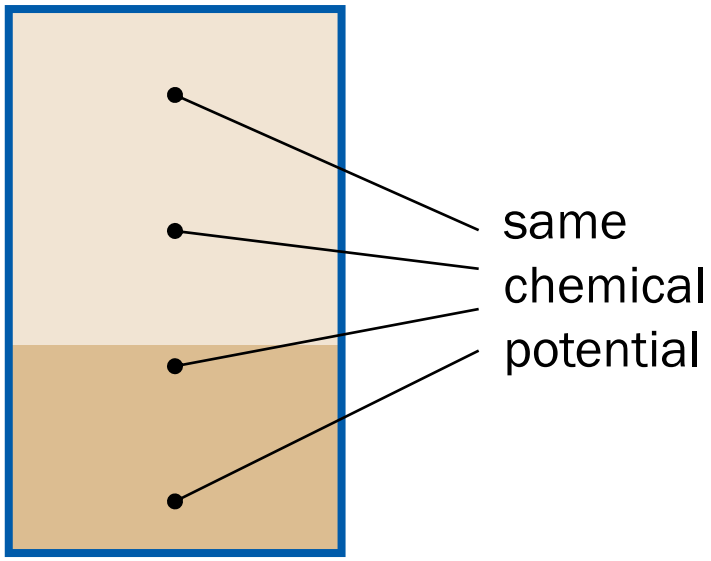
Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_{\mathrm{A}}, n_{\mathrm{B}}, \cdots)$

- Thermodynamic potentials
  - $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$
  - $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$
  - $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$
  - $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$

Thermodynamics of Phases

At equilibrium, the chemical potential of a substance is the same in and throughout every phase in the system



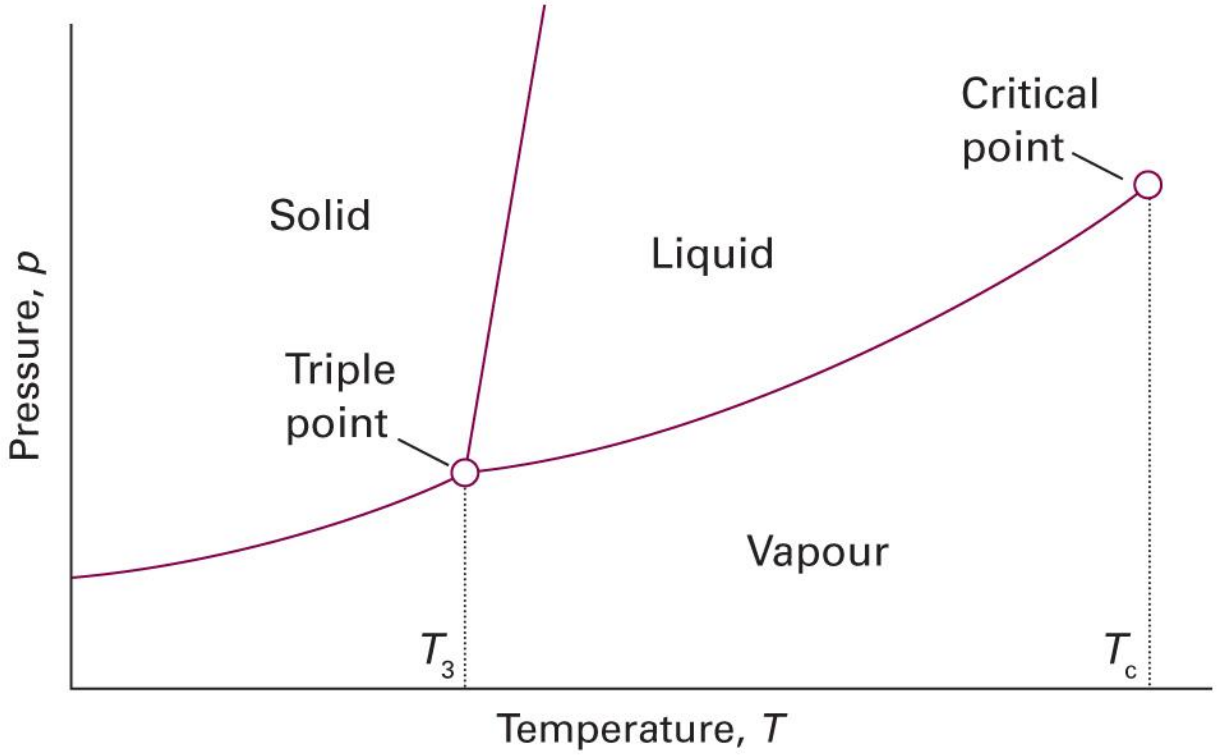
Phase

A form of matter that is uniform throughout

in chemical composition and physical state

- Phase diagram: A diagram that shows the regions where its various phases are thermodynamically stable

Typical Phase Diagram



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.4)

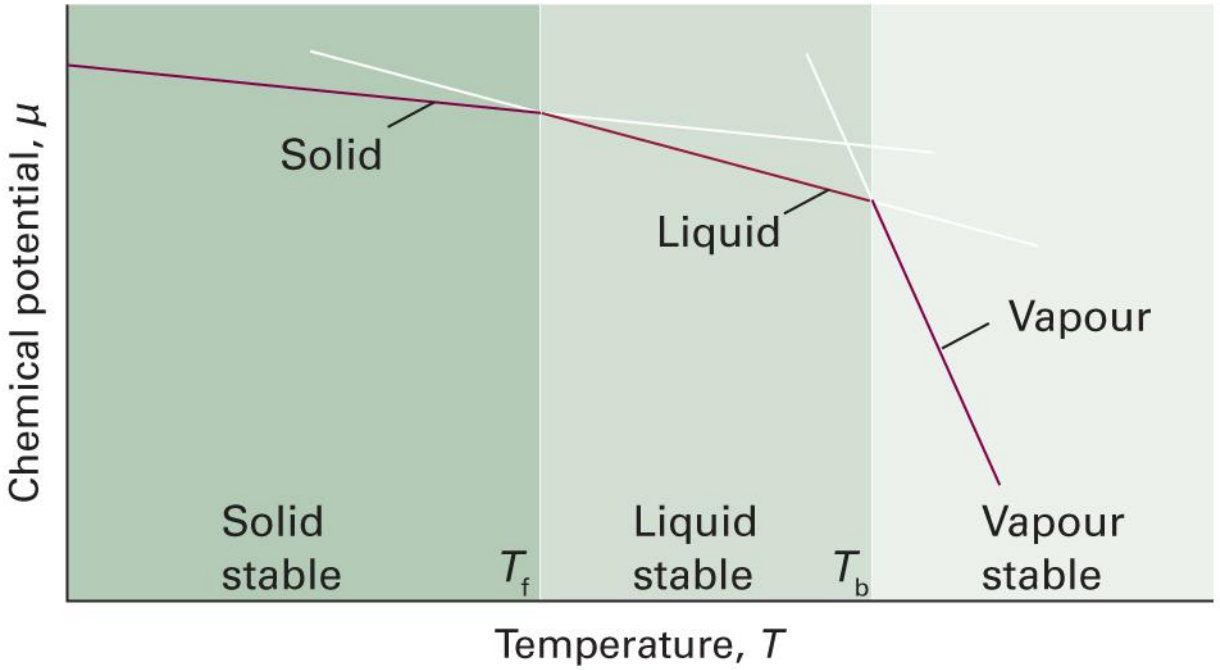
Phase Transition

Spontaneous conversion

of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs}, the two phases are at equilibrium

Chemical Potential



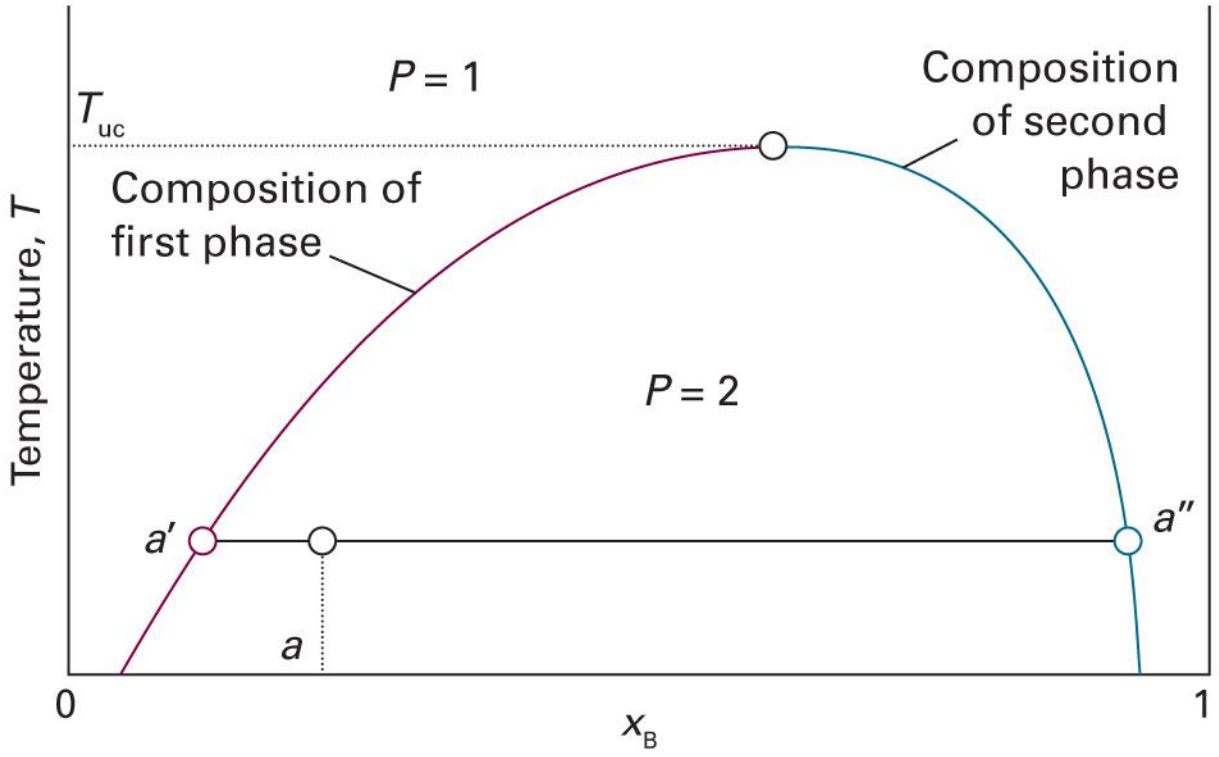
(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figures 4B.1)

Multi-Component System

- Possible phases
  - Each component: gas, liquid, solid
  - Mixed/demixed

- *e.g.* two-component system (binary system)
  - A: liquid, B: liquid, A-B: mixed
  - A: liquid, B: liquid, A-B: demixed
  - A: gas, B: gas, A-B: mixed
  - …

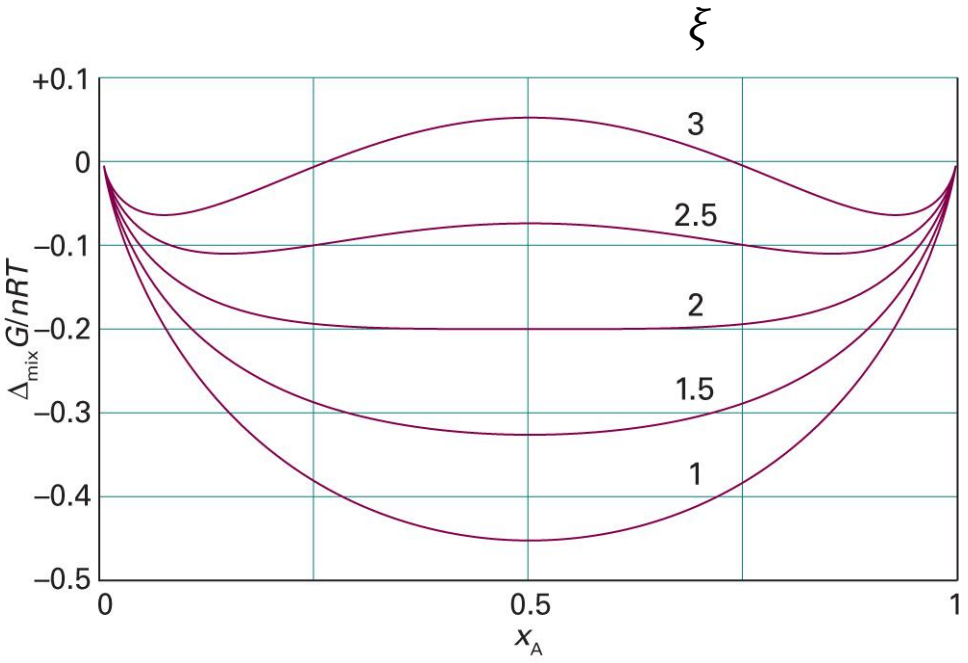
Example: Binary System



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.14)

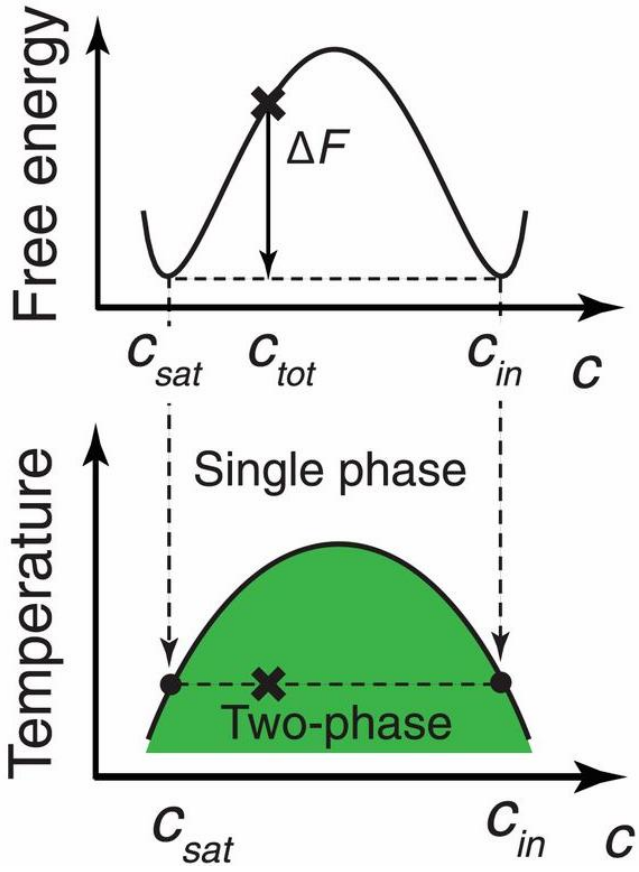
Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}} + \xi x_{\text{A}} x_{\text{B}})$$



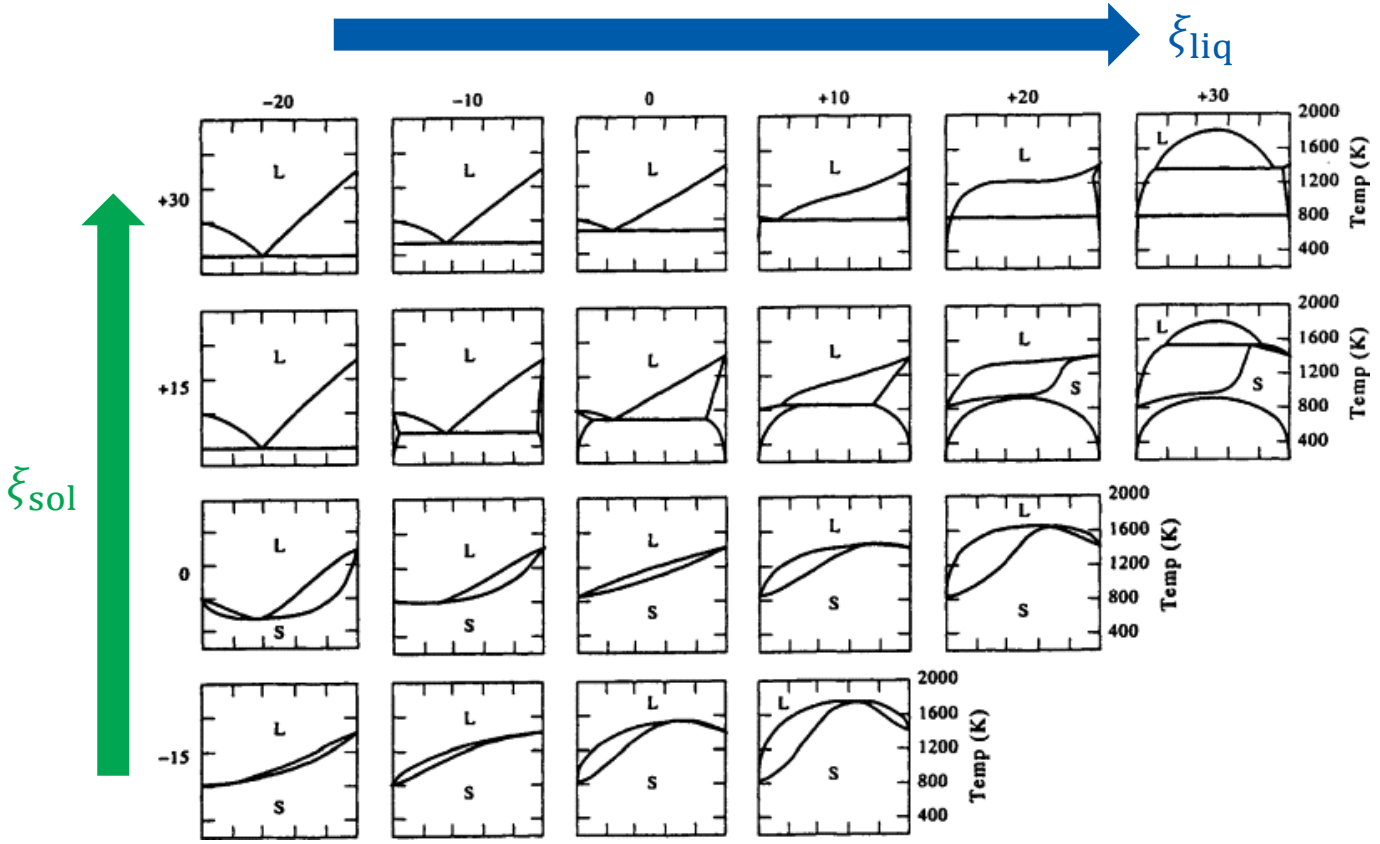
(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Liquid-Solid Binary Mixture

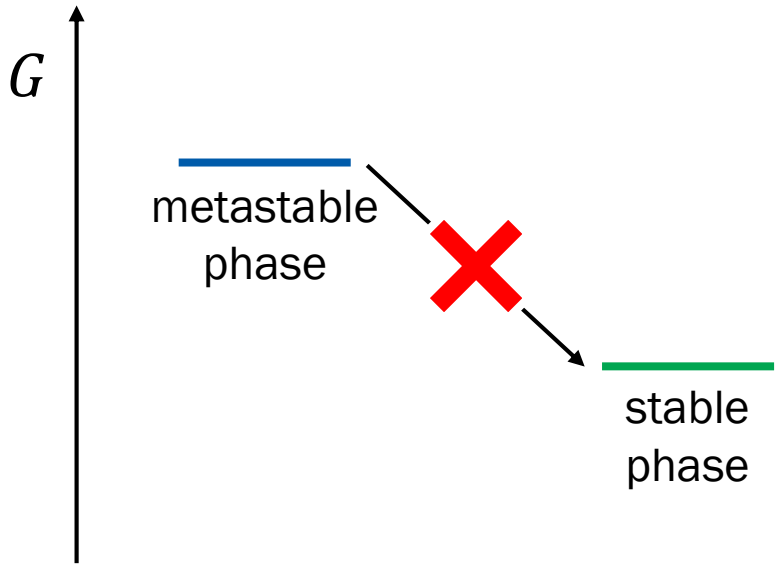


(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

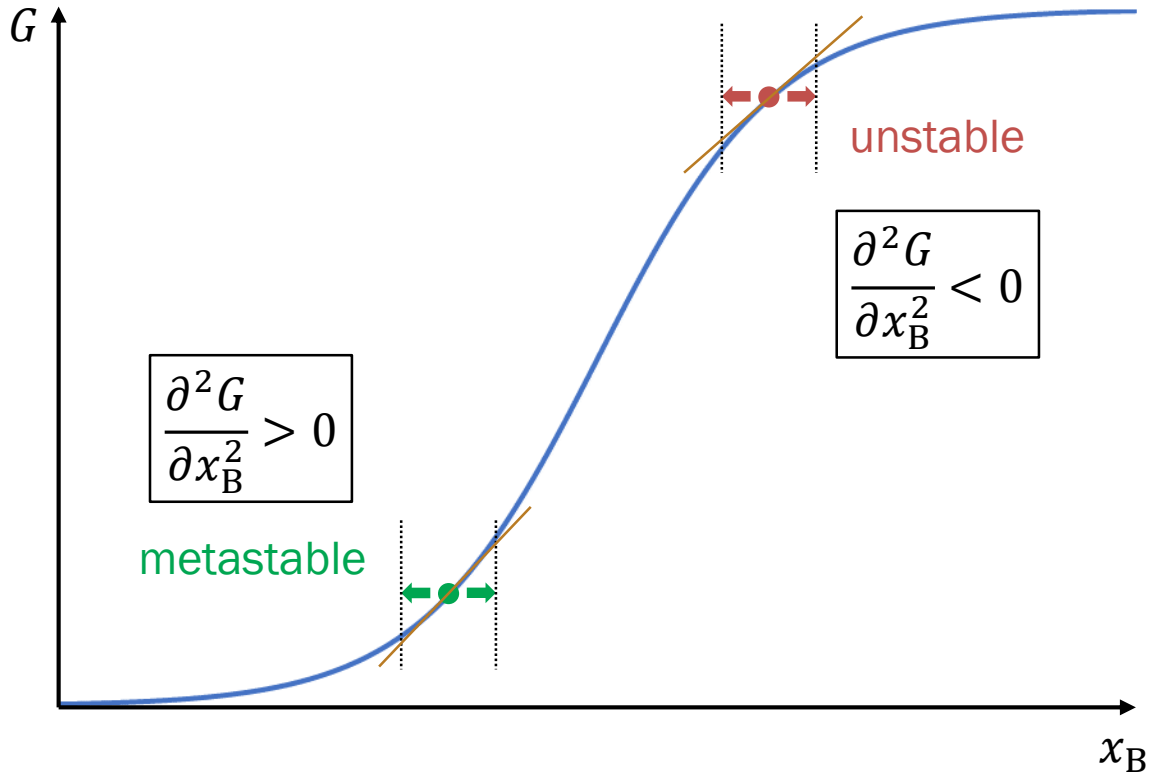
Metastable Phase

Thermodynamically unstable phases that persist

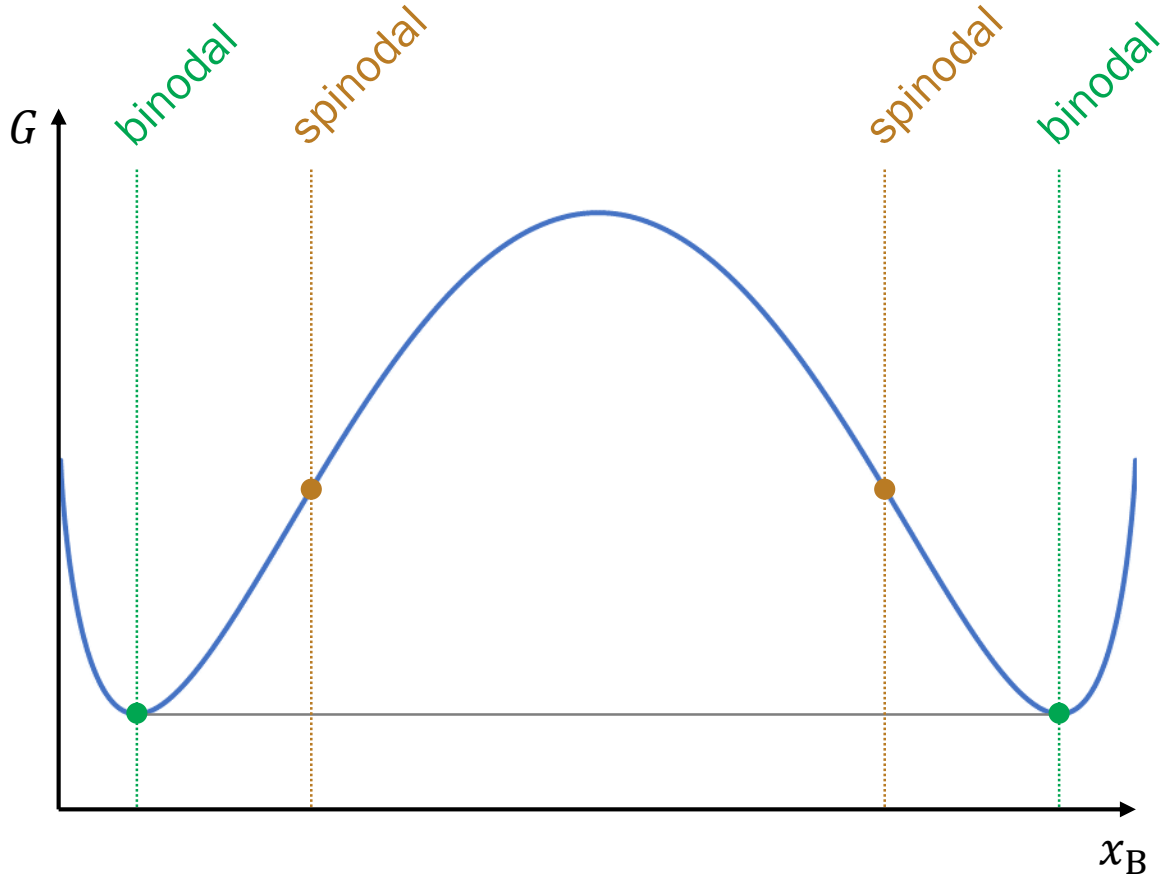
because the transition is kinetically hindered



Metastable? Unstable?



Binodal and Spinodal



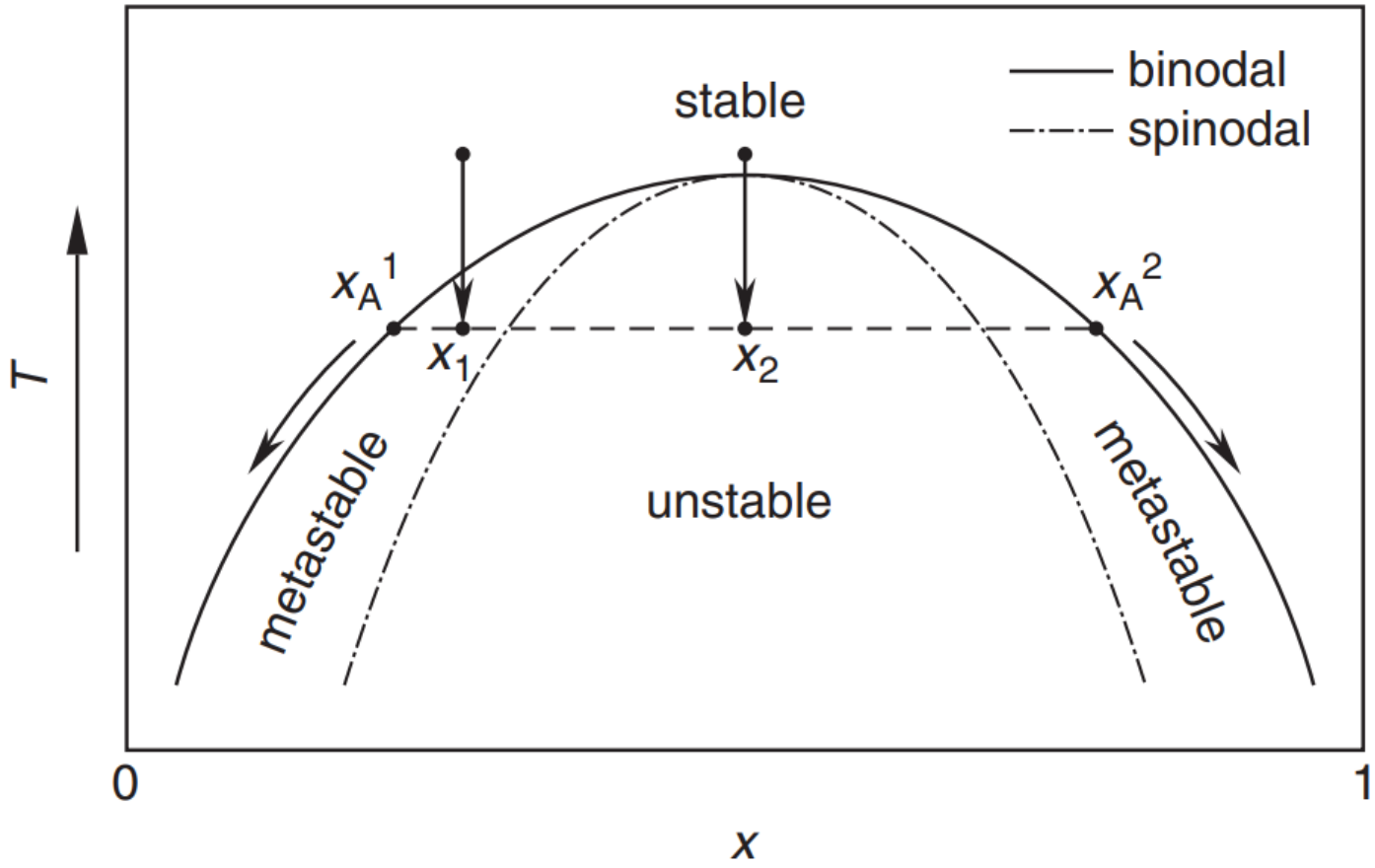
Binodal:

$$\frac{\partial G}{\partial x_B} = 0$$

Spinodal:

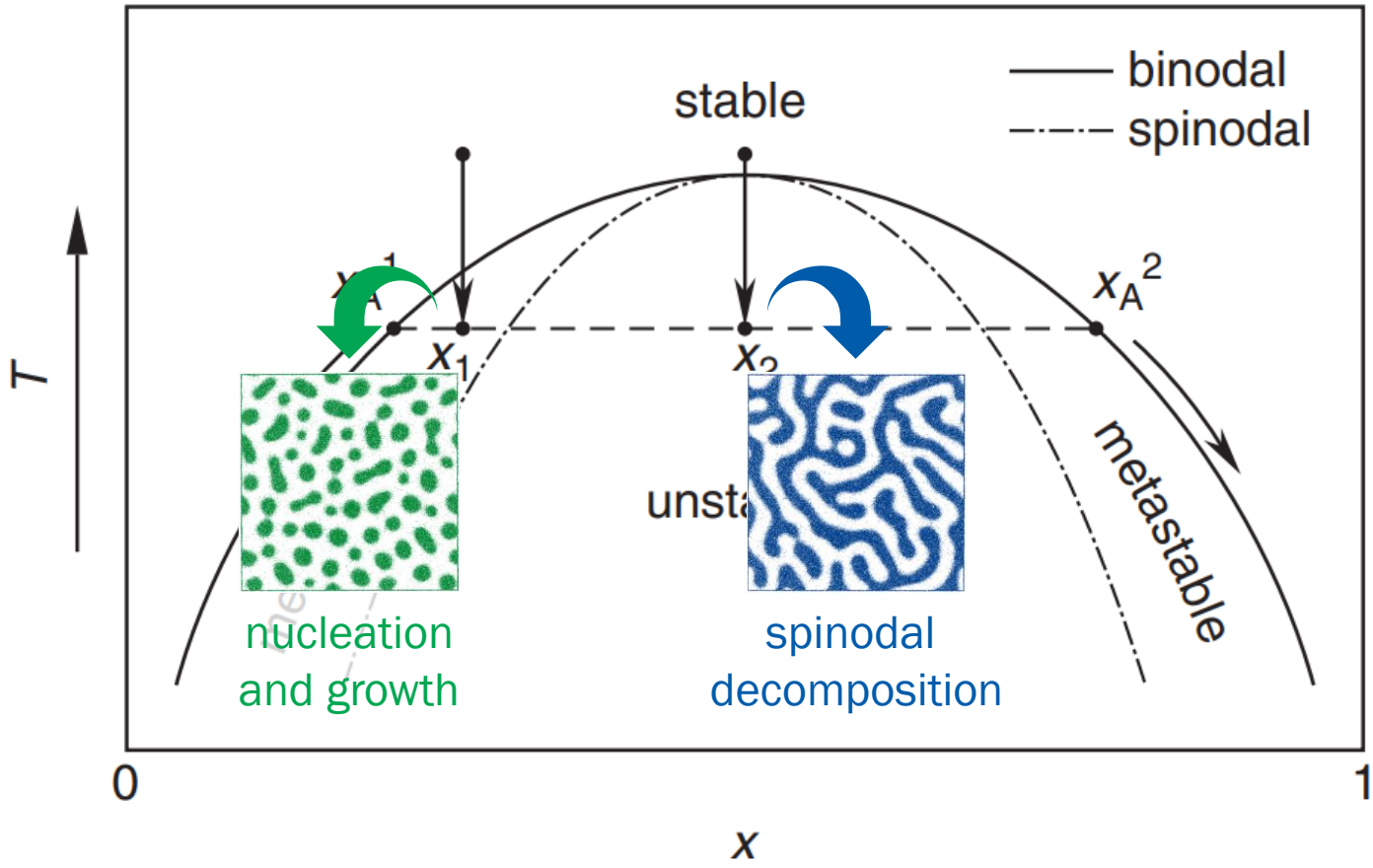
$$\frac{\partial^2 G}{\partial x_B^2} = 0$$

Two Lines on Phase Diagram



(Image: Cheng, ***Phase Transitions in Polymers*** (2008), Fig. 2.14)

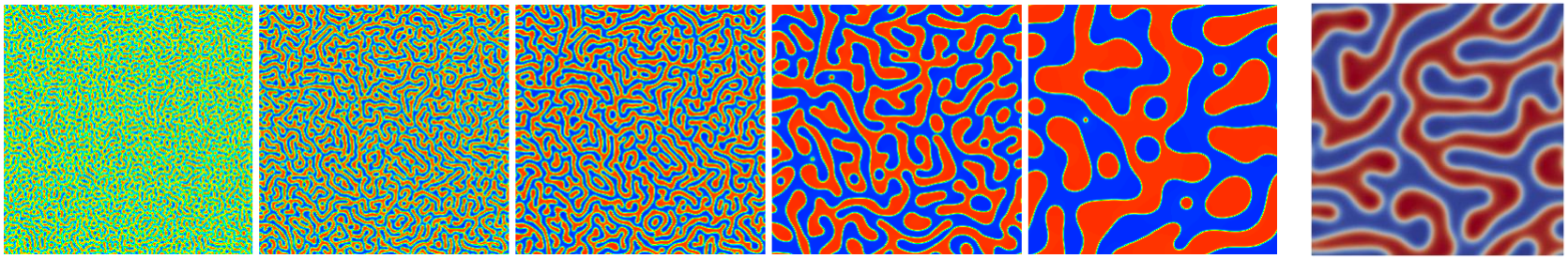
Kinetics of Phase Separation



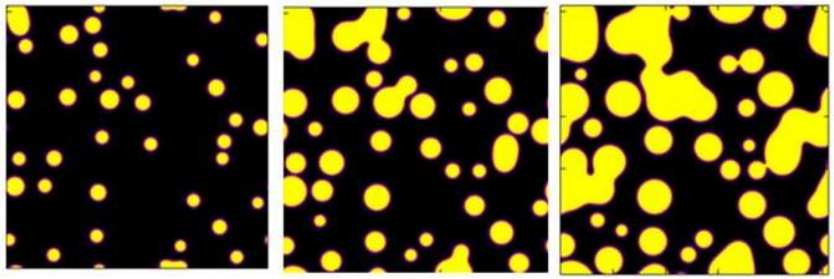
(Images: Cheng, ***Phase Transitions in Polymers*** (2008), Fig. 2.14 / Gebauer *et al.*, ***Chem Soc Rev*** (2014), Fig. 3)

Kinetics of Phase Separation

- Spinodal decomposition



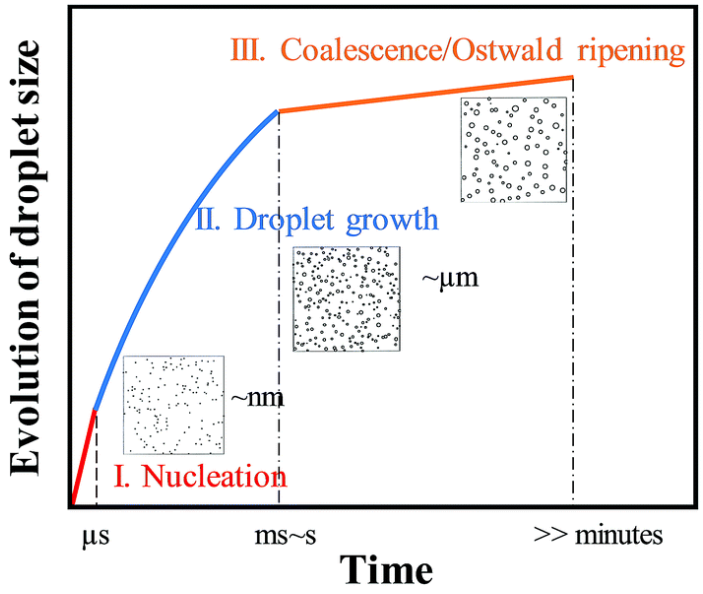
- Nucleation and growth



(Images: https://doi.org/10.2316/P.2013.804-040
https://fuckphasefield.com/spinodal-decomposition-in-2d/
https://doi.org/10.1007/s11837-014-1033-9)

Nucleation and Growth

metastable phase ⟶ stable phase



Universal in chemistry and physics!

(Image: Luo *et al.*, ***Phys Chem Chem Phys*** (2017), Fig. 6)

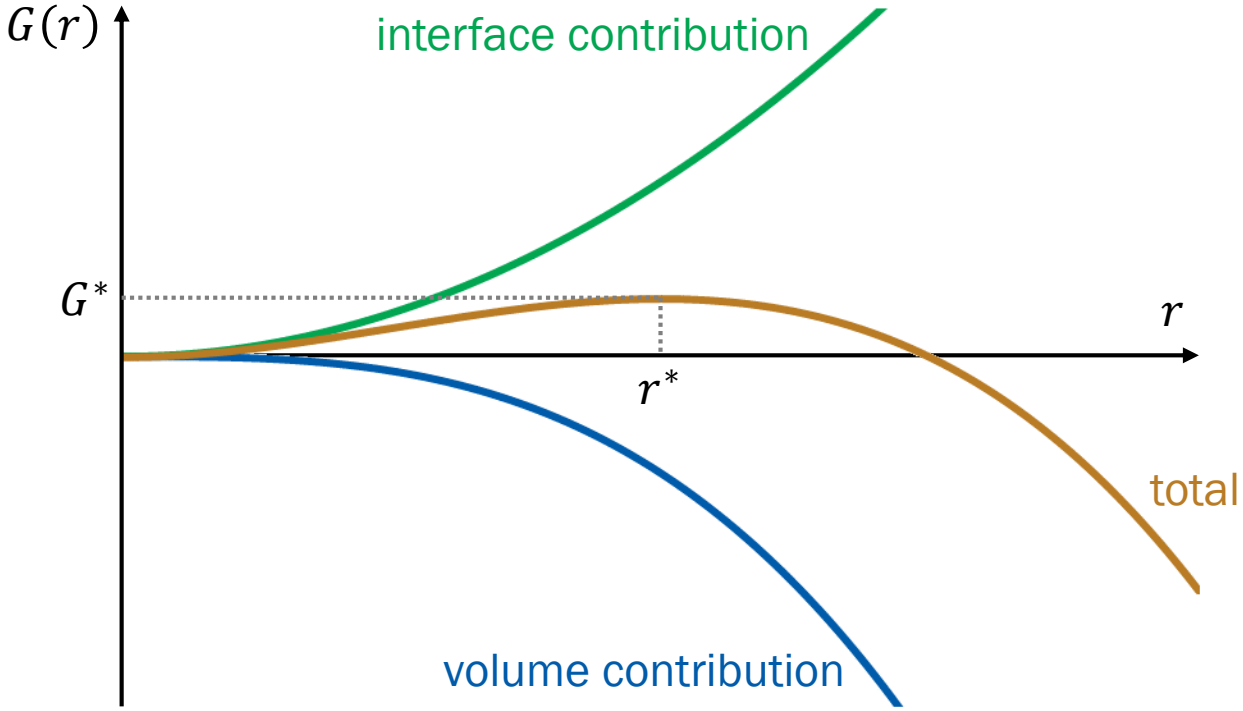
Classical Nucleation Theory

- Gibbs energy density $\Delta g = g_{\text{stable}} - g_{\text{metastable}} < 0$
- Surface tension $\gamma > 0$

- Spherical droplet of size r
- Gibbs energy of the droplet

$$G(r) = V\Delta g + A\gamma = \frac{4}{3}\pi r^3 \Delta g + 4\pi r^2 \gamma$$

Gibbs Energy of Droplet



Critical Size

$$G(r) = \frac{4}{3}\pi r^3 \Delta g + 4\pi r^2 \gamma$$

- Critical size r^*

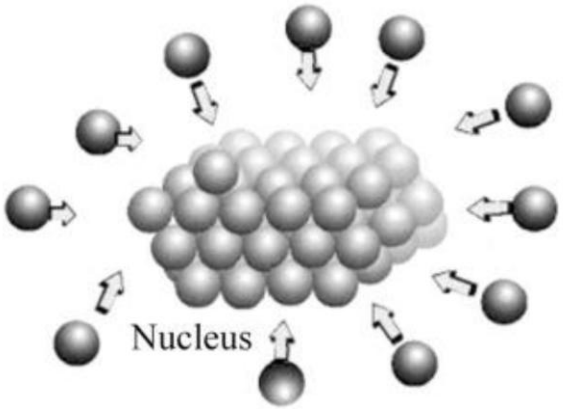
$$\frac{\mathrm{d}G(r)}{\mathrm{d}r} = 4\pi r^2 \Delta g + 8\pi r \gamma = 0$$

$$r^* = -2\gamma/\Delta g \text{ (nontrivial)}$$

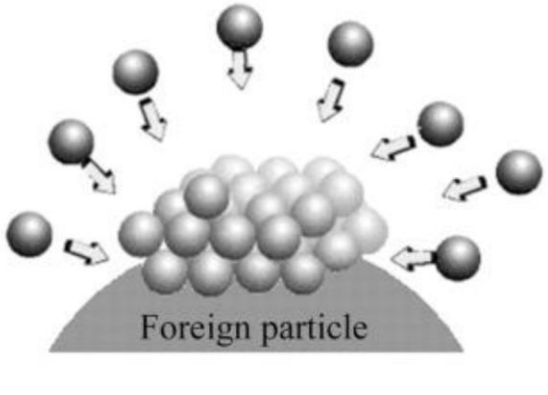
- Activation energy G^*

$$G^* = G(r^*) = \frac{4}{3}\pi\left(-\frac{2\gamma}{\Delta g}\right)^3 \Delta g + 4\pi\left(-\frac{2\gamma}{\Delta g}\right)^2 \gamma = \frac{16\pi\gamma^3}{3(\Delta g)^2}$$

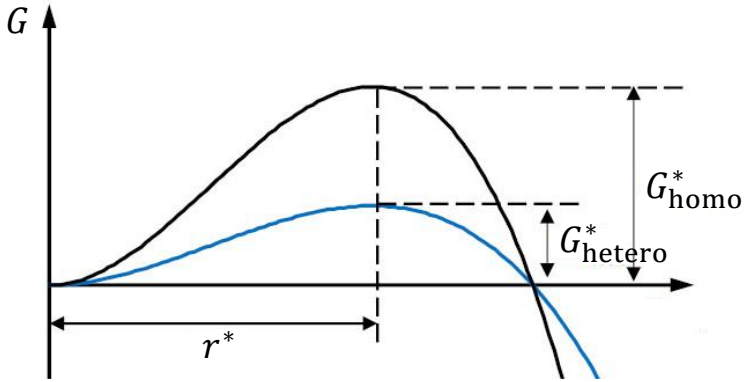
Homogeneous vs. Heterogeneous



Homogeneous nucleation

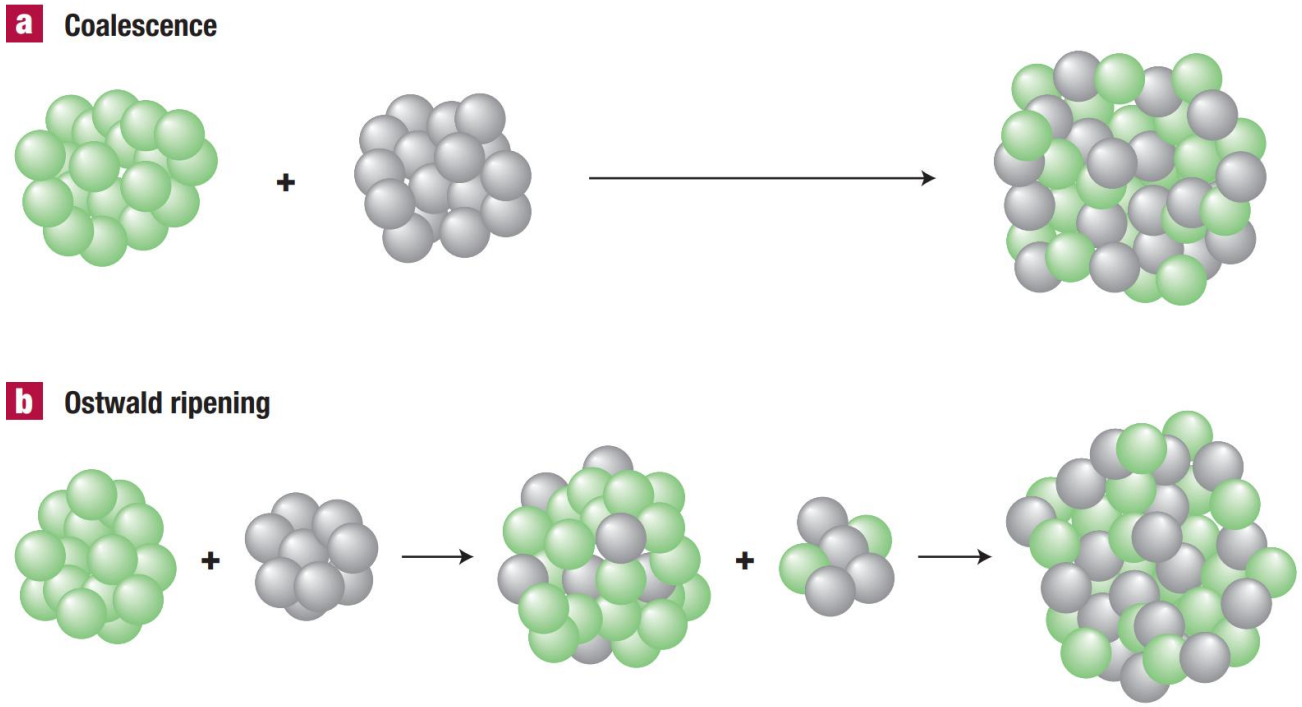


Heterogeneous nucleation



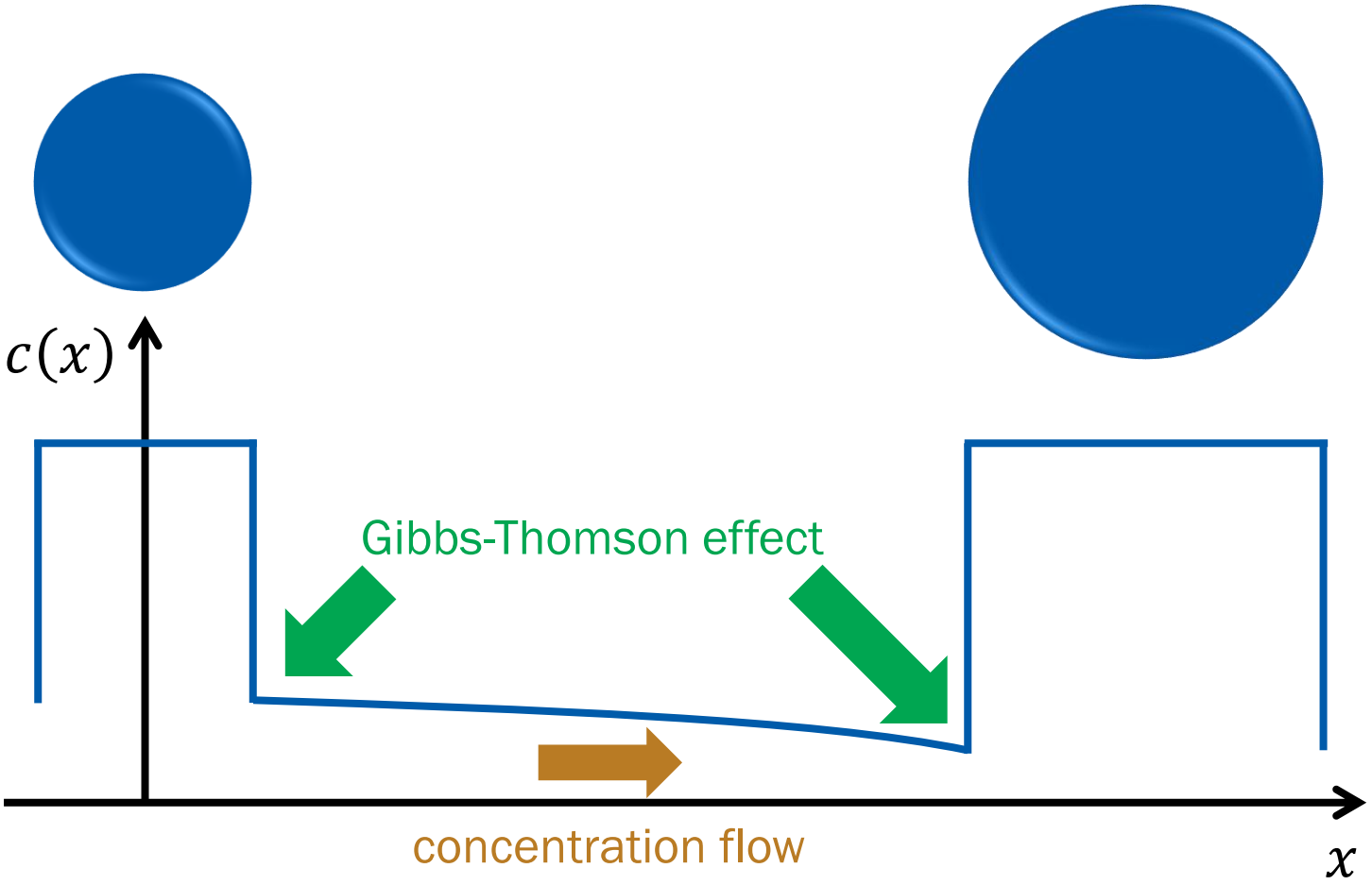
(Images: Chen *et al.*, ***Sustainability*** (2022), Fig. 7 / https://www.elementum3d.com/white-papers/understanding-solidification-and-nucleation-in-aluminum-alloys/)

Late Stage of Growth



(Image: Bowker, *Nature Materials* (2002), Fig. 1)

Ostwald Ripening

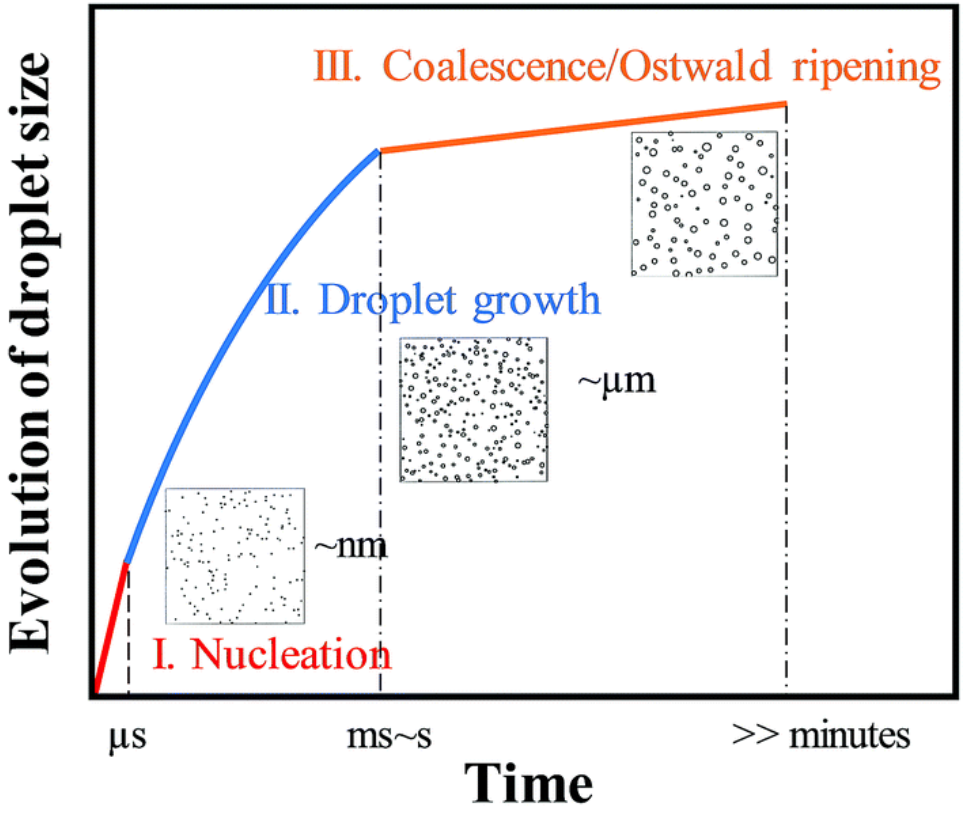


Ostwald Ripening



(Image: Huang *et al.*, ***Nature Communications*** (2017), Supp. Movie 1)

Nucleation and Growth



(Image: Luo *et al.*, ***Phys Chem Chem Phys*** (2017), Fig. 6)